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Deep eutectic solvent-driven mild lignocellulose pretreatment: Unlocking lignin valorization and carbohydrate digestibility

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ABSTRACT

High-value utilization of lignocellulose and its components is generally constrained by the pretreatment efficiency and the structural robustness of fractionated components. Harsh pretreatment conditions can promote pretreatment efficiency but also cause serious condensation of lignin structure, thus inhibiting the depolymerization of lignin into valuable chemicals. Herein, we designed a novel deep eutectic solvent (DES) composed of pyridine hydrochloride, ethylene glycol, and $AlCl_3$ for the pretreatment of lignocellulose. In the developed DES, the lignocellulose could be efficiently fractionated into the natural lignin derivatives with satisfactory hydrogenolysis reactivity and the carbohydrate fraction with high enzymatic digestibility owing to the synthesized DES having adjustable hydrogen-bond acidity, high affinity for lignin, and good protection role on lignin structure. After being pretreated under a mild fractionation condition (70°C, 6 h), a promising delignification ratio of 70.4 % was obtained for poplar, and the extracted lignin exhibited native substructure with well-preserved β-O-4 linkages (42.1 per 100 aromatic units), abundant hydroxyl groups content (4.97 mmol/g), and high molecular weight (5643 g/mol). These structural features contributed to the conversion of the obtained lignin into aromatic monomers with a high yield (27.4 %). Besides, the pretreated residue could be efficiently converted into glucose with a yield of 88.2 % due to the abundant hydroxyl groups of lignin in the cellulose-rich residues. Remarkably, comprehensive studies of lignin structural evolution during pretreatment revealed that the structure-controllable lignin could be effectively isolated by modulating the hydrogen-bond acidity of DESs. This work proposed a mild and efficient lignin-first biorefinery approach that was conducive to the integrated valorization of lignocellulose.

1. Introduction

Utilization of lignocellulosic biomass as a renewable energy resource is a promising alternative to reduce current dependence on petrochemical resources. Nevertheless, lignocellulose is a rigid matrix of cellulose, hemicellulose and lignin intertwined through an abundance of covalent and hydrogen bonds, which complicates processing it into fuels, chemicals, and functional materials [1,2]. Fractionation of lignocellulose, which involves deconstruction of plant cell walls, is important for the complete and efficient utilization of these three main constituents [3]. During lignocellulose fractionation, the reaction intensity (high temperature, high pressure, high acidity) tends to be directly proportional to the delignification ratio and inversely proportional to the lignin activity and cellulose retention [4]. Efficient delignification generally involves harsh pretreatment conditions, and the resulting high-purity cellulose has many applications in enzymatic saccharification and the production of cellulose nanofiber [5,6]. However, irreversible condensation under harsh environments is a commonly used procedure that severely obstructs the facile, effective, and selective transformation of lignin [7]. Thus, developing efficient delignification methods without seriously affecting the structural integrity of lignin is a critical issue in the co-valorization of carbohydrates and lignin.

In order to achieve efficient valorization of lignin and carbohydrates, lignin-first biorefinery strategies have been proposed to protectively fractionate active lignin while facilitating delignification [8–10]. These strategies include mild organosolv pretreatment [11–13], ammonia-

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based pretreatment [14,15], and ionic liquid fractionation [16]. The success of these methods was attributed to the use of protective media and mild processing conditions, which allowed the extent of cleaving β-aryl ether bonds and forming reactive benzylic intermediates to be diminished, thus facilitating the preservation of β-O-4 linkages and decreasing the condensation ratio. Alternatively, deep eutectic solvents (DESs) are promising green solvents with highly favorable properties, including low vapor pressure, non-volatile, low cost, easy preparation, and tunable properties [17–19]. Owing to the tunable properties of DESs, the structural integrity of fractionated lignin can be ensured by selecting protective reagents as DESs components [20]. For example, Barta's group showed that the introduction of ethylene glycol (EG) into the choline chloride/oxalic acid DES could protect the β-O-4 linkages in birch lignin during pretreatment at 100°C for 24 h, which eventually resulted in a significant improvement in monophenol and glucose yields [21]. Xu et al. reported that pretreatment of bagasse with choline chloride/glyoxylic acid monohydrate DESs at 100°C for 9 h could preserve high β-O-4 content (48.3/100 Ar) lignin at a delignification ratio of 68 % [22]. Bian's group observed that 37.5 % delignification of corncob was achieved by using ethylene glycol/oxalic acid DES at 70°C for 6 h and the regenerated lignin maintained extremely high β-O-4 substructures (56.2/100 Ar) [23]. Despite the promising performance of DESs pretreatment in lignin extraction with structural integrity, the corresponding low delignification ratios and relatively harsh conditions (high temperature, and/or long time) were still unsatisfactory.

To achieve good co-valorization of lignin and carbohydrates, novel DESs which can promote delignification efficiency while maintaining good preservation of lignin structural integrity under mild pretreatment conditions is highly desired. As has been noted, the uneven distribution of electron density in N-heterocyclic compounds can strongly affect their interactions with electron-dense molecules [24]. Lignin is an aromatic polymer with a higher electron density than cellulose and hemicellulose, which results in lignin having a unique affinity for N-heterocyclic compounds. Lin et al. demonstrated that the pyrazole-lactic acid DES displayed a high affinity to lignin and could selectively extract lignin from wheat straw with a yield of up to 80 % and a β-O-4 content of up to 39/100 Ar [25]. Inspired by these results, a novel DES (denoted PEA-DESs) composed of pyridine hydrochloride (PH), EG, and $AlCl_3$ were constructed for mild pretreatment of various lignocelluloses with the purpose of regenerating as much lignin with structural integrity as possible. We hypothesized that the acidic environment provided by $AlCl_3$ during PEA-DESs pretreatment would effectively deconstruct lignocellulose, while PH would enhance the affinity for lignin thus accelerating lignin dissociation, and the subsequently dissociated lignin would be protected by EG through nucleophilic addition. Initially, a comprehensive study was carried out to evaluate the delignification efficiency of the novel DES on different lignocelluloses as well as the structural integrity of the extracted lignin. Subsequently, the evolution of lignin during the pretreatment process of DESs with different hydrogen-bond acidity was investigated in detail, and the possible mechanisms of fractionation and lignin protection were proposed. Finally, the specific high-value conversion of the pretreated fractions was conducted.

2. Materials and methods

2.1. Materials and reagents

The lignocellulosic biomasses (poplar, pine, bagasse) used in this study were sourced from an experimental field in Beijing, China. These biomasses were processed into powders and subsequently sieved using 50-mesh nets. Before being used in the pretreatment, each lignocellulosic biomass was pre-extracted with toluene/ethanol (2:1, v/v) to remove wax. The cellulase (Cellic CTec2, 100 FPU/mL) was obtained from Novozymes Investment Co., Ltd., China. Pyridine hydrochloride (98 %) and ethylene glycol (99 %) were purchased from Macklin Chemical Co., Ltd (Shanghai, China), and aluminum chloride (98 %) was supplied by Aladdin Biochemical Co., Ltd (Shanghai, China). All commercial chemicals were used directly without further purification.

2.2. Preparation and characterization of PEA-DESs

The desired PEA-DESs were prepared by heating a mixture of PH, EG, and $AlCl_3$ at the desired molar ratio at 70°C until a homogeneous transparent liquid was produced. The Kamlet-Taft solvatochromic parameters (α, β, and π^*, quantifying hydrogen-bond acidity, hydrogen-bond basicity, and dipolarity/polarizability, respectively), which can be used to characterize the chemical properties of DESs [26], were determined according to the procedure reported previously [27], and the relevant formulas are listed in the Supporting Information.

2.3. Lignocellulose pretreatment and lignin fractionation with PEA-DESs

In a typical preparation, 3 g of lignocellulose was added to 45 g of the prepared DES in a sealed flask and heated at designated temperatures (e.g., 70, 80, and 90°C) for 6 h under continuous stirring. Then the mixture was cooled and diluted with 100 mL anhydrous ethanol, and the cellulose-rich residues were recovered by filtration (washed with extra 200 mL EtOH) and dried in a desiccator. The ethanol was evaporated from the filtrate by rotary evaporation to obtain lignin-rich DES fraction. Subsequently, the concentrated filtrate was precipitated in dilute HCl (pH = 2) and filtered to obtain lignin. The cellulolytic enzyme lignin (CEL) was extracted from lignocellulose (poplar, pine and bagasse) as described in the Supporting Information. The compositions of the raw and pretreated lignocellulose were determined according to the National Renewable Energy Laboratory standard analytical method (NREL/TP-510–42618) [28]. The delignification ratio and lignin yield were calculated as follows:

$$Delignification\ ratio = (m_1 - m_2)/m_1 \times 100\% \quad (1)$$

$$Lignin\ yield = m_3/m_1 \times 100\% \quad (2)$$

where m_1 and m_2 were the mass of lignin in the raw lignocellulose and pretreated residue (g), respectively. m_3 was the mass of regenerated lignin.

2.4. Enzymatic hydrolysis of reactive residues from the DESs pretreatment

In a typical preparation, the obtained residue (200 mg), sodium acetate buffer (50 mM, pH 4.8, 10 mL) and cellulase preparation (15 FPU/g) were added into a stoppered Erlenmeyer flask. Then, enzymatic hydrolysis was conducted at 50°C in an incubator shaker at 150 rpm for 72 h. During the enzymatic hydrolysis, 100 μL aliquots were taken at different incubation times and the amount of glucose generated was determined using high-performance liquid chromatography (HPLC). The glucose yield was calculated from:

$$Glucose\ yield = \frac{glucose\ in\ the\ hydrolysate\ (g) \times 0.9}{cellulose\ in\ the\ solid\ residue\ (g)} \times 100\% \quad (3)$$

2.5. Structural characterizations of extracted lignin

FT-IR spectra of lignin were recorded in the wavenumber range of 4000–600 cm^{-1} on a Bruker Tensor II spectrometer (Bruker, Germany) using the potassium bromide pellet technique.

All NMR analyses of lignin were recorded on a Bruker Avance 500 MHz spectrometer (Bruker, Germany) and the results were processed in Bruker TopSpin 4.1.3 software. For the 2D heteronuclear single-quantum coherence (HSQC) NMR analysis, 50 mg of lignin was dissolved in 0.5 mL of DMSO-d_6. The spectral widths were 5,000 and 20,000 Hz for the ^1H and ^{13}C dimensions, respectively. A total of 1024 points were collected for the ^1H dimension with a recycle delay of 1.5 s,

while 64 or 256 transients were used for the ^{13}C dimension. Identification and semiquantitative calculations were performed according to the literature [21,29,30], and are described in the Supporting Information.

The quantitative ^{31}P NMR analysis was performed after phosphitylation with 2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholane in pyridine/$CDCl_3$ (1.6/1, v/v) is described elsewhere [31].

The number-average and weight-average molecular weights (M_n and M_w, respectively) of lignin were estimated with gel permeation chromatography (GPC) on an Agilent 1200 (Agilent, Japan). All lignin samples were fully acetylated in acetic anhydride/pyridine before GPC analysis, and the lignin acetylation was performed as reported previously [30]. The specific lignin acetylation process is described in the Supporting Information. Acetylated lignin (4 mg) was dissolved in 2 mL of HPLC-grade tetrahydrofuran (THF) and filtered through 0.22 μm polytetrafluoroethylene filters. The M_w value of lignin was calculated using a polystyrene standard sample, and the GPC curve was analysed using Peakfit software.

2.6. Valorization of PEA-extracted lignin

2.6.1. De-etherification of PEA-extracted lignin

In a typical procedure, 200 mg of extracted lignin was dissolved in 5 mL 1,4-dioxane/water (2:1, vol/vol) containing 0.1 M HCl and heated in a sealed pressure-resistant tube at 100°C for 5 h. Subsequently, the lignin solution was added dropwise to 50 mL of deionized water to regenerate the de-etherified lignin, which was collected by filtration and freeze dried.

2.6.2. Catalytic hydrogenolysis of the PEA-extracted lignin

The catalytic hydrogenolysis of lignin has been described in the literature [32]. In a representative preparation, 50 mg of the extracted lignin, 5 mL methanol and 50 mg Ru/C were placed into a 50 mL Parr autoclave (Parr Instrument Company). The reactor was sealed and purged with H_2 three times to expel air, then charged with 3 MPa H_2. The reaction was conducted at 200 °C with a stirring speed of 400 rpm for 12h. After the completion of the reaction, the insoluble fraction was removed by filtration, and lignin oily products were obtained from the soluble fraction by rotary evaporation. Tetradecane, as internal standard, was added to the lignin oily solution in anhydrous THF and stirred for 10 min. The resulting solution was filtered through a 0.22 μm PTFE filter and subjected to GCMS on a Shimadzu GCMS-QP2010SE (Shimadzu, Japan) equipped with an HP-5 MS column and MS Shimadzu GC-2010 equipped with an HP-5 column to identify and quantify the components. The total yield of all the generated monomers was calculated as follows:

$$Yield_{monomer}(wt\%) = M_{monomer}/(20\ mg \times Purity) \times 100 \quad (4)$$

where $M_{monomer}$ and *Purity* were the mass of the monomers and the

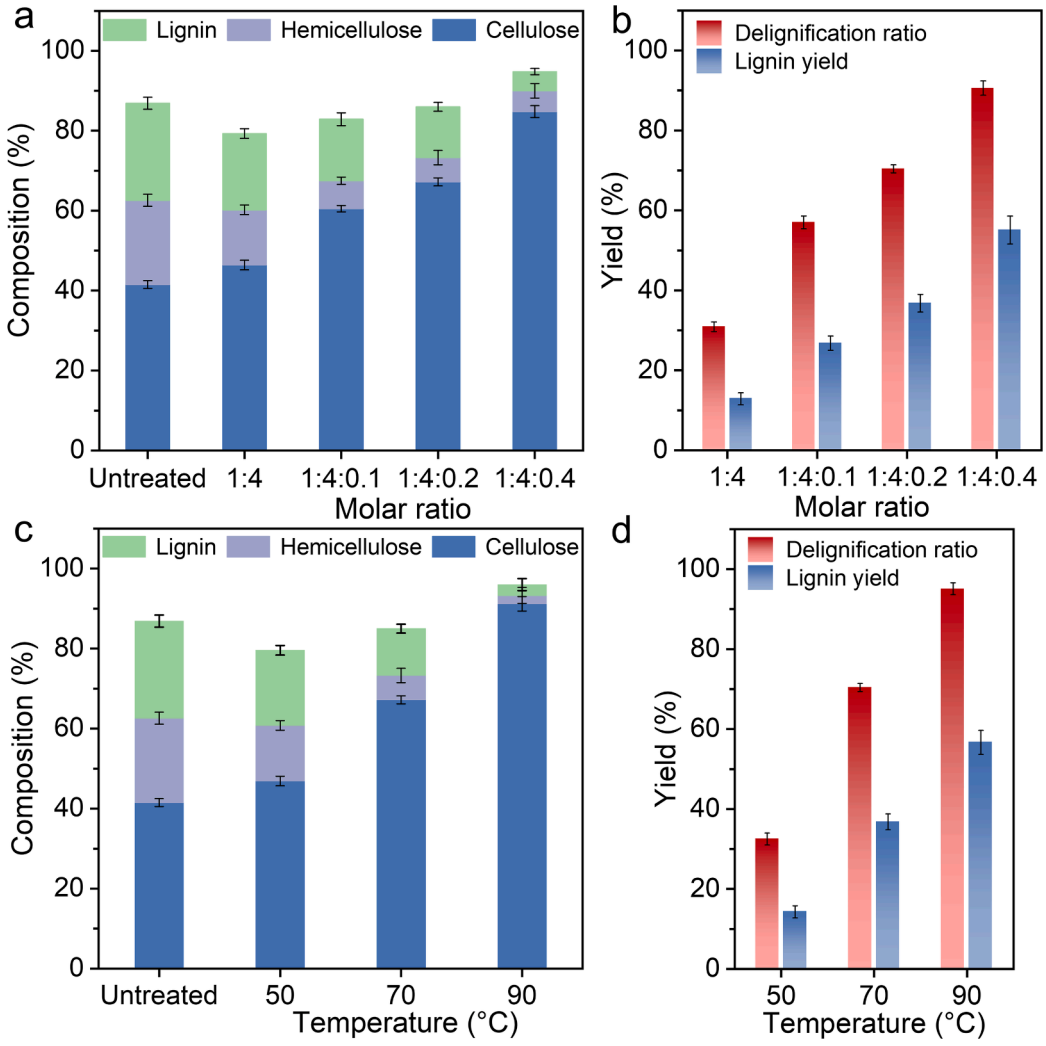


Fig. 1. Pretreatment of poplar by PEA-DESs. Effect of $AlCl_3$ molar content (a, and b) at 70 °C, and influence of different temperatures (c and d) at the molar ratio of 1:4:0.2 in the PEA-DESs.

purity of the corresponding lignin, respectively.

3. Results and discussion

3.1. Evaluation of delignification efficiency and lignin yields

Generally, the pretreatment efficiency of poplar was closely related to the molar ratio of components in the DESs and the employed pretreatment temperature. Therefore, a detailed optimization of the reaction conditions was performed and analyzed (Fig. 1 and Table S1), including $AlCl_3$ molar contents (0, 0.1, 0.2, and 0.4) and temperatures (50, 70, and 90°C). Initially, the effect of the $AlCl_3$ molar content in DESs was investigated at 70°C. With an increase of $AlCl_3$ content from 0 to 0.4, the hemicellulose and lignin recovery decreased from 21.1 % to 5.2 % and 24.3 % to 4.8 % (Fig. 1a), respectively, and the delignification ratio increased significantly from 30.9 % to 90.6 % (Fig. 1b). These results demonstrated that increasing $AlCl_3$ content promoted lignocellulose fractionation. Probably, the number and strength of electrostatic/anionic hydrogen-bond interactions between Cl^- with lignin increased with increasing of $AlCl_3$ content, thus facilitating lignin removal [33]. The Kamlet-Taft solvatochromic parameters (α, β and π^*) were able to effectively reflect chemical properties of the DESs, and the parameters for the prepared DESs were determined and are tabulated in Table S2. As reported, DESs possessing higher α value exhibited better lignin dissociation capacity [34]. Obviously, the α value of PEA-DESs was most significantly affected by the $AlCl_3$ content. The α value increased from 1.17 to 1.75 when the $AlCl_3$ molar content increased linearly from 0 to 0.4 (α value $= 1.43n_{AlCl_3} + 1.2$, R^2: 0.97) (Fig. S1a). Remarkably, the delignification ratio showed a linear correlation (delignification ratio $= 100.6\alpha$-83.9, R^2: 0.97) with the α value of PEA-DESs (Fig. S1b). Therefore, the addition of $AlCl_3$ as a Lewis acid could effectively enhance the hydrogen bonding acidity of DESs, resulting in easy donation of H-bonds to liberate H^+, which could promote the cleavage of the lignin-carbohydrate complex (LCC) linkages and β-aryl ether bonds to dissociate lignin [35,36]. Moreover, $AlCl_3$ competed with pyridine hydrochloride for the hydroxyl groups of ethylene glycol. Thus, more and stronger anionic hydrogen bonds were formed to release the pyridine cation with unevenly distributed electron density, which could further increase the affinity of DESs to the lignin with high electron density, thereby facilitating delignification [25].

In summary, the excellent performance of PEA-DESs on low-temperature pretreatment originated from the synergistic effects of enhanced hydrogen-bond acidity by $AlCl_3$ and the improved affinity of pyridine hydrochloride to lignin. Subsequently, the effect of temperature on pretreatment efficiency was studied using the PEA-DES with a molar ratio of 1:4:0.2 (Fig. 1c-d). The delignification ratio increased from 32.5 % to 95.1 % as the temperature was increased from 50 to 90°C. This was primarily because higher pretreatment temperatures enhanced the deconstruction of plant cell walls, enabling the cleavage of covalent and hydrogen bonds in LCC and lignin macromolecules by PEA-DESs. Notably, the lignin yield was lower than the delignification rate (Fig. 1b, d), which might result from the excessive depolymerization of lignin and structural functionalization during pretreatment [23].

3.2. Structural elucidation of PEA-extracted lignin

To disclose the structural dissimilarity between lignin extracted by PEA-DESs-(1:4:0.2) at 70°C ($PEA_{0.2}$-L) and cellulolytic enzyme lignin (CEL), 2D HSQC NMR, FTIR, ^{31}P NMR, and GPC analysis of both types of lignin were performed. First, structural information on the number of basic aromatic units and interlinkages as well as the condensation ratio of lignin was comprehensively investigated by 2D HSQC NMR (Fig. 2). The identification of the main lignin cross-peaks in the HSQC spectra was conducted based on previous studies [21,29], and the assignments are given in Table S3. In the side-chain region, the typical interlinkages were detected in CEL, including ether bonds (largely β-O-4 motifs, 50.7/

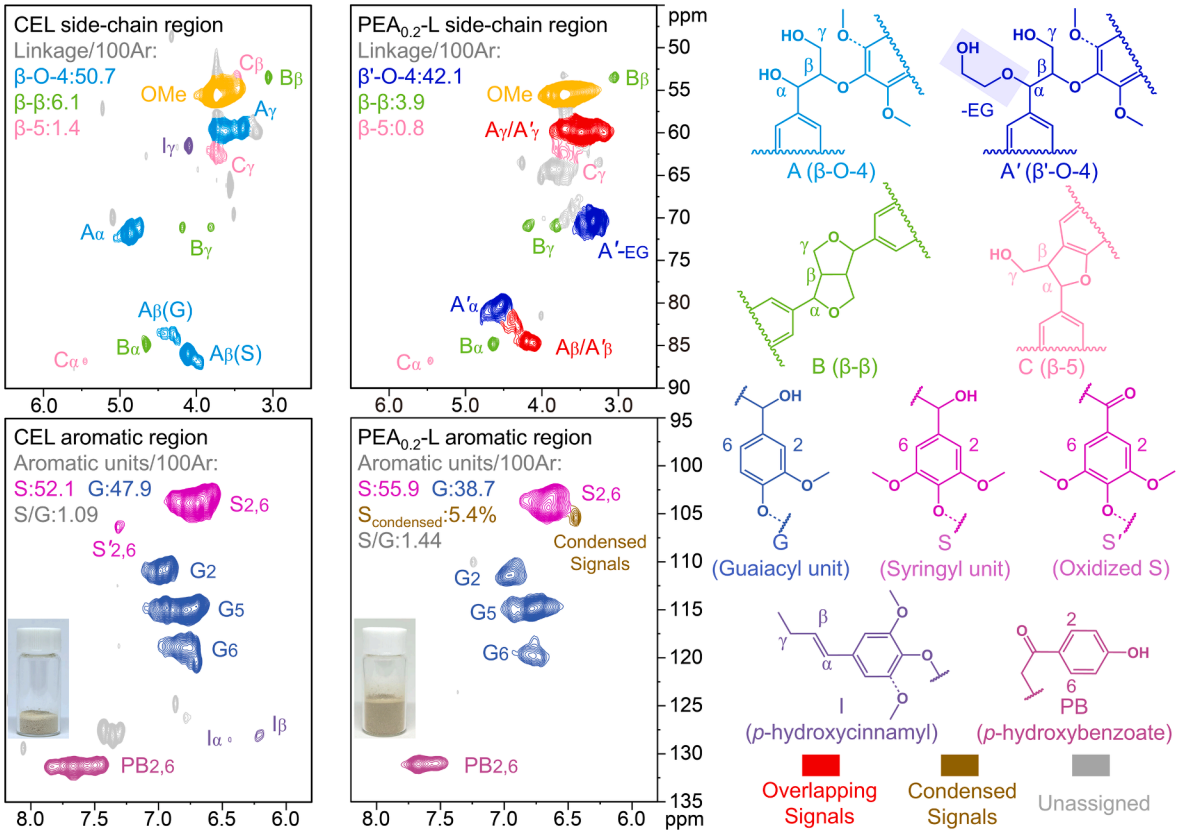


Fig. 2. Elucidation of the interlinkages and aromatic units in CEL and $PEA_{0.2}$-L by 2D HSQC-NMR spectroscopy.

100 Ar) and carbon–carbon interunit bonds (largely β-β and β-5, 6.1 and 1.4/100 Ar, respectively). Meanwhile, the intact syringyl-type (S-type) unit (52.1/100 Ar) and guaiacyl-type (G-type) unit (47.9/100 Ar) signals were exhibited in the aromatic region of CEL (Fig. 2). In comparison, PEA-DESs as an acidic solvent could promote the formation of benzylic intermediates via a dehydration step at the α-OH of β-O-4 substructure [11]. Therefore, the structure of lignin obtained from the PEA-DESs pretreatment was different in comparison with CEL. In particular, the addition reaction of ethylene glycol and active carbocation intermediates could occur in the $PEA_{0.2}$-L. Obviously, in the side-chain spectrum of $PEA_{0.2}$-L, a new cross-peak at δ_C/δ_H of 70.5/3.36 (A'_{EG}) which was assigned to the methylene carbons of EG clearly indicated that EG had been successfully incorporated into the lignin structure. Likewise, the C_α-H_α signal of EG-substituted β-O-4 substructure was identified at δ_C/δ_H of 80.9/4.57 (A′α). Importantly, EG acted as an effective stabilization reagent for lignin functionalization, protecting the β-aryl ether bonds from cleavage and inhibiting subsequent condensation [21,37]. The α-alkoxylated β-O-4 content in $PEA_{0.2}$-L was 42.1/100 Ar, which was similar to that of CEL. However, the signal of the native β-O-4 linkage became invisible, suggesting that $AlCl_3$ was an effective catalyst for the dehydration of α-OH in β-O-4 motifs and the alkoxylation of ethylene glycol with lignin. Furthermore, weak condensation signals (condensed S: 5.4 %) in the spectrum of $PEA_{0.2}$-L were also observed, implying that the incorporation of EG into lignin could block subsequent lignin condensation. In addition, the S/G value in $PEA_{0.2}$-L (1.44) was higher than that in CEL (1.09), and the contents of S-type units increased slightly from 52.1/100 Ar to 55.9/100 Ar while the G-type units decreased significantly from 47.9/100 Ar to 38.7/100 Ar, probably because G-type units of lignin were preferentially degraded over S-type units during PEA-DESs pretreatment. Unlike CEL, the cross-peak of *p*-hydroxybenzoate substructure (PB) was weakened in $PEA_{0.2}$-L, indicating effective disruption of the ester bonds between the PB units and carbohydrate [38,39]. Based on the results above, it was deduced that PEA-DESs-assisted fractionation caused efficient lignin dissociation along with a substantial retention of β-O-4 substructures.

Characteristic functional groups on PEA-DES-extracted lignin were confirmed by FT-IR (Fig. 3a). The lignin extracted by PEA-DESs ($PEA_{0.2}$-L) showed almost all the characteristics absorption bands compared with CEL, including distinct absorption peaks for the aromatic ring (1596, 1506, 1425 cm^{-1}), unconjugated carboxyl group (1708 cm^{-1}), hydroxyl group (3425 cm^{-1}), S-type units (1329, 1124 cm^{-1}), and G-type units (1268, 1225, 1035 cm^{-1}) [40,41], which implied that $PEA_{0.2}$-L preserved the structure of native lignin. Nevertheless, one significant difference was observed. The stretching vibration for the conjugated carbonyl groups detected at 1658 cm^{-1} in the CEL disappeared in the spectra of $PEA_{0.2}$-L. As has been previously reported, the carbonyl group as an oxygen-containing polar unsaturated group readily undergoes addition reactions with oxygen-containing nucleophilic reagents (like EG) [42]. Thus, the disappearance of the carbonyl signals in the $PEA_{0.2}$-L side chain is mainly attributed to the nucleophilic addition of EG in PEA-DESs.

Hydroxyl groups are an important functional group in lignin, which not only increases the hydrophilicity of lignin, but also facilitates various chemical reactions of lignin. The content of different types of hydroxyl groups was determined using ^{31}P NMR (Fig. S2). As shown in Fig. 3b, the content of aliphatic hydroxyl groups in PEA-DESs decreased mildly compared to that in CEL, which was attributed to the conversion of hydroxyl groups to form benzylic carbocation intermediates by the dehydration and acylation in the acidic PEA-DESs [43]. Further, the change in the aliphatic OH group content was also associated with alkoxylation of the dehydrated lignin with EG. Moreover, the content of the phenolic OH group in the PEA-DES extracted lignin ($PEA_{0.2}$-L)

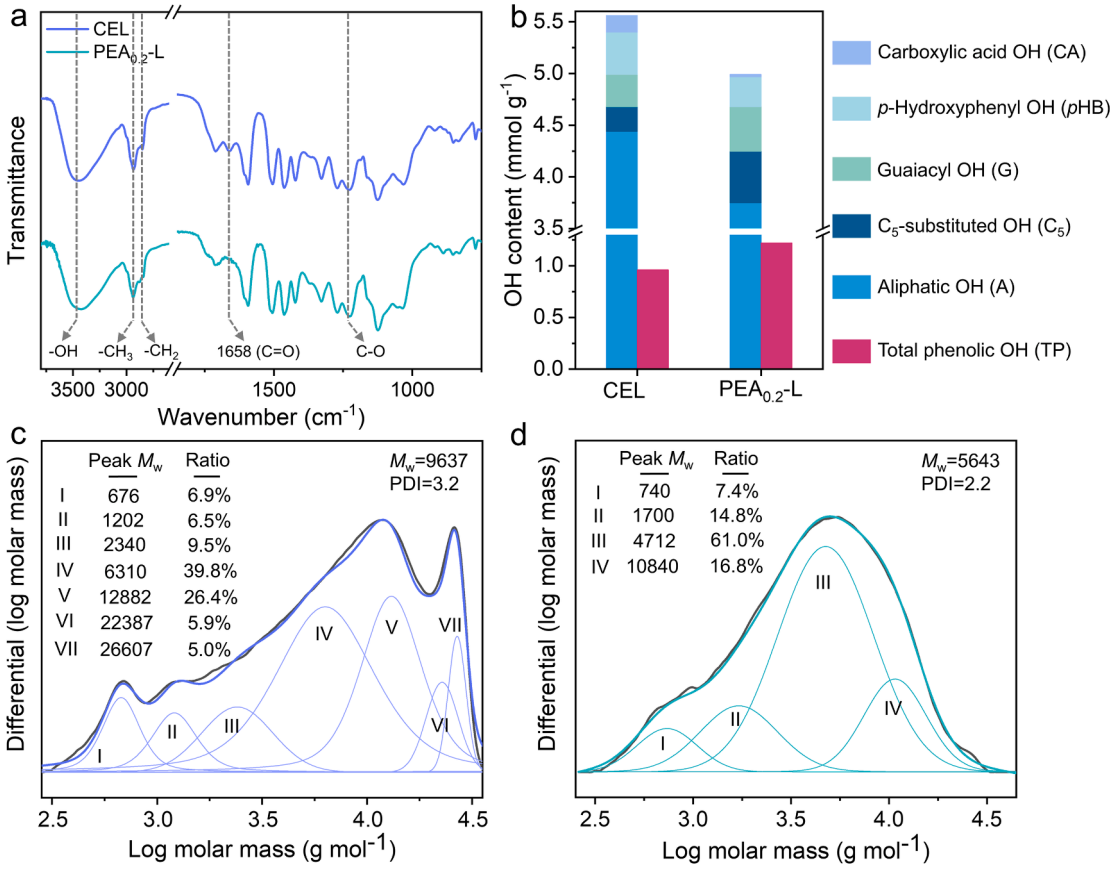


Fig. 3. FTIR spectra of CEL and extracted $PEA_{0.2}$-L (a), quantitation of hydroxyl contents in CEL and $PEA_{0.2}$-L by ^{31}P NMR (b), and GPC-determined molecular weight distributions of CEL (c) and $PEA_{0.2}$-L (d).

increased by 27.1 % compared to that in CEL (0.96mmol g^{-1}), which implied that a small amount of the β-O-4 linkages cleaved during the pretreatment process, thus leading to an increase in the phenolic OH groups. It is worth noting that the increased phenolic hydroxyl group content was mainly derived from C_5-substituted OH groups and to a lesser extent guaiacyl OH groups. Combined with the 2D HSQC NMR results, it was deduced that the cleavage of the G-type β-O-4 bonds during pretreatment dissociated the G-type lignin fragments with phenolic hydroxyl groups, and the C-2 of these G-type lignin fragments formed condensed structures with C-5 in other lignin units, which increased the C_5-substituted OH groups [44]. More importantly, this slight content variation of phenolic and aliphatic OH groups also suggests that the β-O-4 motifs were highly preserved during the PEA-DESs fractionation process.

Log-normal distribution functions were fitted to the GPC curves of the lignin species to separate the peaks and determine the molecular weight distributions. The data and fittings for CEL and $PEA_{0.2}$-L are shown in Fig. 3c and 3d, respectively. The molecular weight distribution of CEL could be fitted to 7 lignin fractions with different M_w values, and the ratio of oligomeric (< 3000 g/mol, including I, II, and III), medium (3000 ~ 10000 g/mol, IV), and supramolecular (> 10000 g/mol, including V, VI, and VII) lignin fractions was calculated to be ~ 22.9:39.8:37.3. In contrast, the GPC curve of $PEA_{0.2}$-L could be modelled by 4 main peaks: (I) 740, (II) 1700, (III) 4712 and (IV) 10840 g/mol. In the molecular weight distribution of $PEA_{0.2}$-L, the medium lignin fractions (III) were predominant (61 %), and the ratio of the supramolecular fractions (IV) dramatically decreased to 16.8 %, while the ratio of the oligomeric fractions (I and II) remained constant (22.2 %). The significantly lower content of the supramolecular lignin fractions in $PEA_{0.2}$-L resulted in its lower M_w (5643 g/mol) and more homogeneous polydispersity index (PDI) (2.2) compared to that for the CEL, which mainly resulted from the cleavage of the lignin *β*-aryl ether bonds during the PEA-DESs pretreatment. Notably, the lignin supramolecular fractions were mainly degraded to the medium fractions, and the content of the oligomeric fractions remained essentially unchanged, implying that lignin linkages were not severely broken during lignin dissociation. In addition, the grafting reaction by EG also increased the molecular weight of the extracted lignin by PEA-DESs. Therefore, the M_w of $PEA_{0.2}$-L was significantly higher than that of the previously reported DESs-fractionated lignin, such as those obtained using choline chloride-glycerol-$AlCl_3$ (1414 g/mol) and choline chloride-*p*-hydroxybenzoic acid (1790 g/mol) [45,46]. It is widely accepted that the molecular weight of lignin is strongly correlated with β-O-4 content, which further confirms that PEA-DESs can extract lignin with a relatively well-preserved structure.

3.3. Universality of PEA-DESs in lignocellulose pretreatment

Inspired by the good performance on the pretreatment of poplar, the applicability of the constructed PEA-DESs on pretreatment of two other types of lignocellulose (pine and bagasse) was investigated under the same conditions (70 °C, 6 h) as those used in the poplar pretreatment. The predominant interlinkages and aromatic units of the fractionated $PEA_{0.2}$-L from pine and bagasse were characterized using 2D HSQC NMR. After the PEA-DES pretreatment, hemicellulose was significantly removed from both pine and bagasse, which decreased from 19.5 % and 26.5 % to 6.5 % and 5.9 %, respectively (Fig. 4a and Table S1). Remarkably, an exceptionally high delignification ratio of 90.4 % was obtained for bagasse, thus providing a very competitive approach to producing cellulose nanofibers with high purity. Softwoods are generally recognized as highly resistant to delignification [47]. Similarly, the PEA-DESs provided a lower delignification ratio (45.5 %) of pine compared to poplar (82.2 %) and bagasse (90.4 %) under the same conditions (Fig. 4b). From 2D HSQC NMR analyses, both lignin

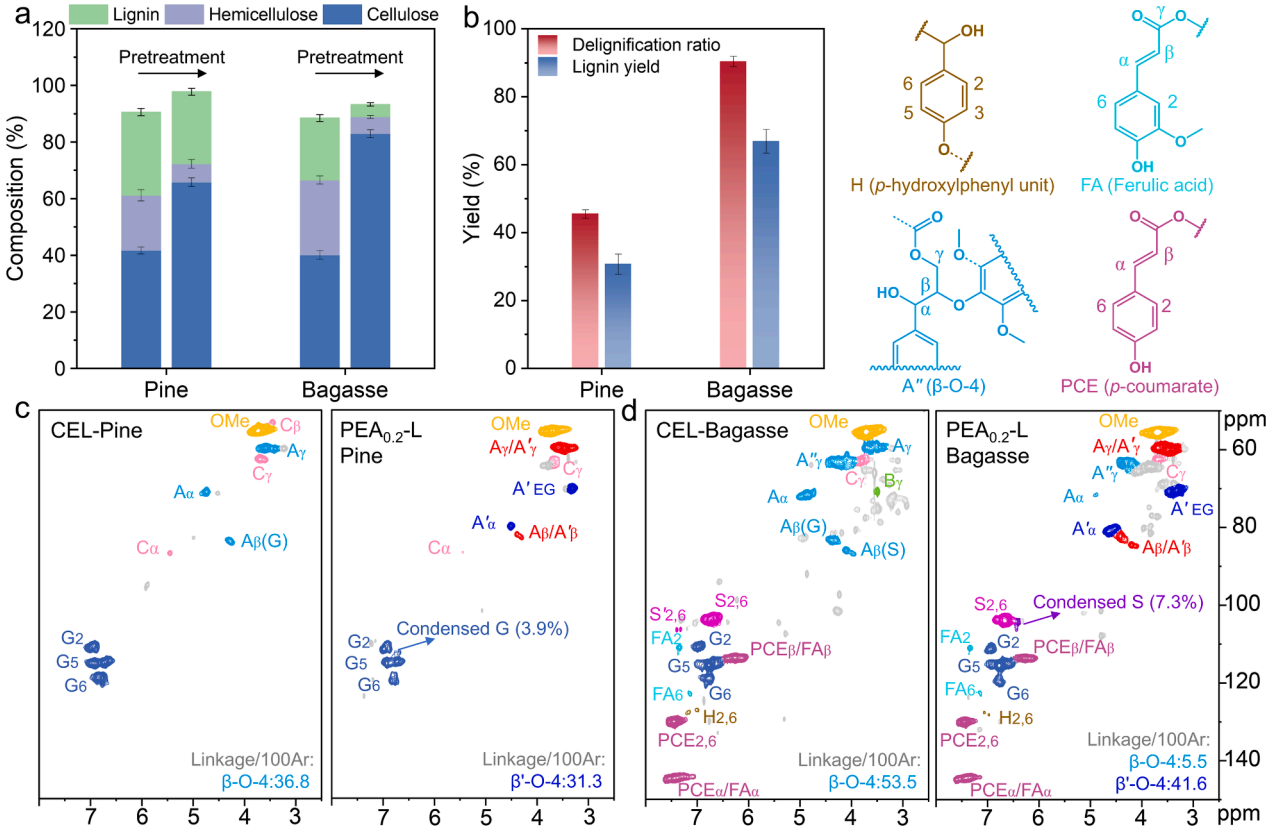


Fig. 4. Compositional content of different lignocelluloses (pine and bagasse) before and after pretreatment (a), the corresponding delignification ratios and lignin yields (b), and 2D HSQC NMR characterization of CEL and $PEA_{0.2}$-L from different lignocelluloses (c, and d).

fractionated from pine and bagasse showed similar structural properties to the $PEA_{0.2}$-L extracted from poplar, i.e., well-preserved β-O-4 substructure and extremely low condensation ratio (Fig. 4c, d, Table S3). In the side-chain region, new α-alkoxylated β-O-4 signals also appeared in the lignin generated from both pine and bagasse, implying the successful incorporation of EG. The yield of $PEA_{0.2}$-L-pine was about 30.7 %, and it retained a high β-O-4 content of 31.3/100 Ar. Meanwhile, a high yield (66.9 %) of $PEA_{0.2}$-L-bagasse was obtained with a content of 47.1/100 Ar for total β-O-4 linkages, of which the contents of native and α-alkoxylated β-O-4 substructures were 5.5 and 41.6/100 Ar, respectively. The contents of native β-O-4 linkages in CEL-pine and CEL-bagasse were 36.8 and 53.5/100 Ar, respectively, indicating that nearly 85 % and 88 % of the β-aryl ether linkages were retained after the PEA-DESs pretreatment. In the aromatic region, the G-type units signal of $PEA_{0.2}$-L-pine exhibited an extremely low condensation ratio (3.9 %). In contrast, the $PEA_{0.2}$-L-bagasse had low amounts of condensed S (7.3 %) and no obvious condensation signals of G-type units.

More interestingly, it was observed that the developed PH-EG-$AlCl_3$ DESs showed much better performance (milder conditions, more efficient delignification, and high structural integrity of lignin) toward some reported DES systems (with the ability to protect β-aryl ether bonds) for the pretreatment of various lignocellulosic biomasses (Table S4). Particularly, the low temperature (70°C) in the PEA-DESs pretreatment system provided a reliable basis for mild and efficient extraction of lignin with natural structures.

3.4. Structural evolution of lignin during PEA-DESs fractionation

As previously reported, DESs with a high α value promote the deconstruction of dense lignocellulose. Pleasingly, the delignification ratio exhibited a linear correlation with the α value of the prepared PEA-DESs (Fig. S1b). However, the DESs with too higher α value cause severe cleavage of β-aryl ether bonds as well as the formation of new C-C bonds by electrophilic aromatic substitution by benzylic carbocation intermediates on the electron-rich aromatic rings, resulting in the loss of active oxygen-containing functional groups (*e.g.,* aliphatic hydroxyl, ether) and natural macromolecular properties of lignin [48,49]. Thus, further investigation on structural and chemical evolution of lignin during PEA-DESs pretreatment was needed for designing suitable solvents to maximize lignin extraction and simultaneously preserve the virgin structure of lignin, which was important to improve lignin quality and better integrate lignin into downstream applications. The evolution of the hydroxyl group content, molecular weight distribution, and the contents of basic structural units and interunit bonding patterns were the crucial factors for evaluating the pretreatment process.

Poplar pretreatment by PEA-DESs with different $AlCl_3$ molar contents (0, 0.1, 0.2, and 0.4) was conducted to dissociate lignin with different structures at 70°C. 2D-HSQC NMR spectra were preformed to provide an insight into basic aromatic units and interlinkages changes of the four lignin samples obtained from the pretreatment by different PEA-DESs with progressively increasing hydrogen bonding acidity. In the linkage region, new signals at δ_C/δ_H of 81.2/4.55 (A'_α) and 70.5/3.36 (A'_{EG}) ppm were observed in all the lignin samples (Fig. 5a-c), confirming that this developed PEA-DESs system was indeed suitable for the

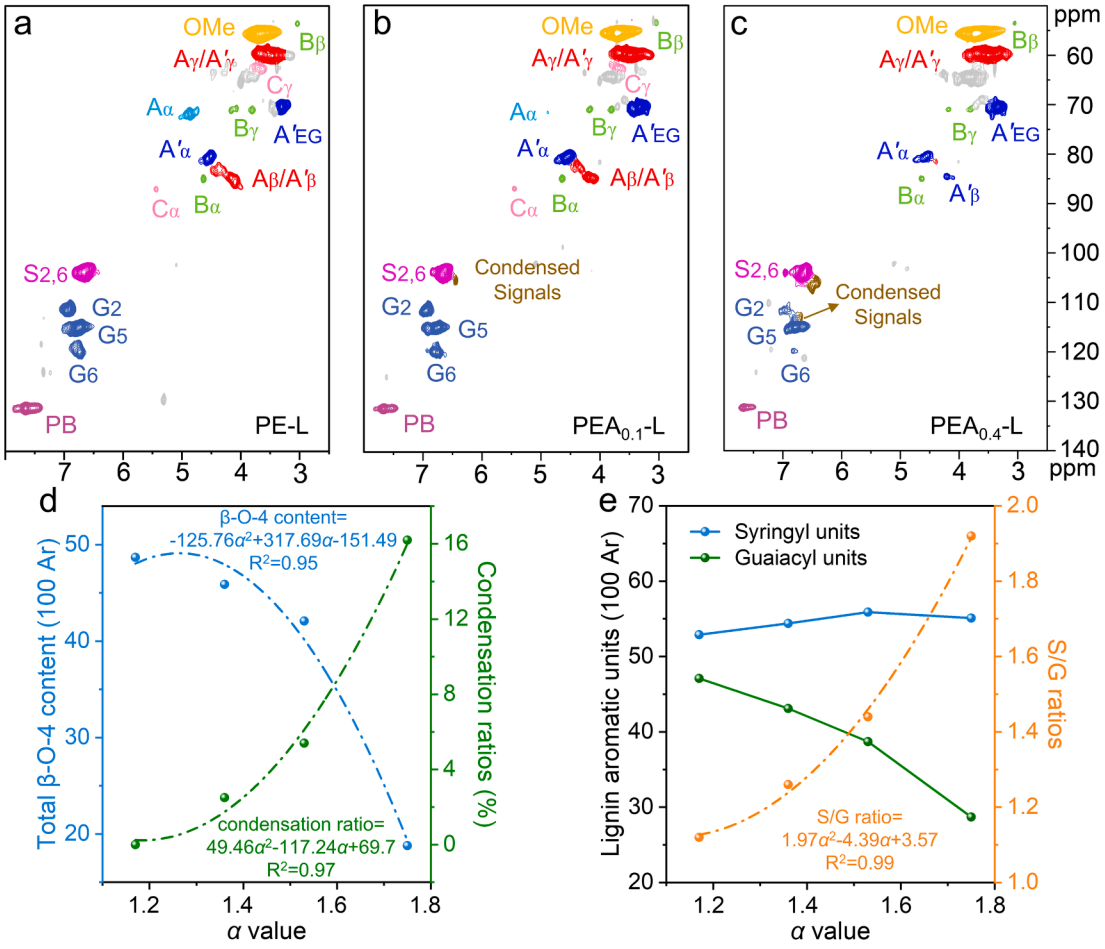


Fig. 5. 2D HSQC NMR spectra of different lignin fractions including PE-L (a), $PEA_{0.1}$-L (b), and $PEA_{0.4}$-L (c), and correlation between the α value of PEA-DESs and total β-O-4 content/condensation ratio (d), and S/G ratio (e). Statistical analysis showed that lignin structures were flexibly regulated based on α value of PEA-DESs.

preservation of aryl ether linkages. The content of aryl ether linkages decreased slowly from 48.7/100 Ar to 42.1/100 Ar as the α value of PEA-DESs increased from 1.17 to 1.53. Meanwhile, the condensation ratio slightly increased from 0 to 5.4 % (Fig. 5d and Table S5). However, when the α value further increased to 1.75, the content of the β-O-4 substructure decreased sharply to 18.8/100 Ar and more obvious signals for condensation (16.2 %) appeared in the aromatic region. Statistical analysis showed that the α value of PEA-DESs had strong quadratic-dependence on with the total β-O-4 content ($\beta-O-4\ content = -125.76\alpha^2 + 317.69\alpha - 151.49$, R^2: 0.95) and condensation ratio (condensation ratio $= 49.46\alpha^2$-$117.24\alpha + 69.7$, R^2: 0.97) (Fig. 5d). Generally, lignin condensation mainly originated from the S-type units. The condensed G-type units were not present in lignin extracted by PEA-DESs with moderate hydrogen bond acidity ($\alpha < 1.53$), and only negligible condensed G-type units were observed in $PEA_{0.4}$-L (Fig. 5a-c). This suggests that the S-type units readily condensed during the lignin fractionation compared to the G-type units [22]. Furthermore, the S/G ratio increased gradually with the increase of α value (S/G ratio $= 1.97\alpha^2$-$4.39\alpha + 3.57$, R^2: 0.99) (Fig. 5e). Specifically, the four extracted lignins showed that the S-type units remained almost unchanged, whereas the G-type units gradually decreased from 47.1 % (PE-L) to 28.7 % ($PEA_{0.4}$-L) (Fig. 5e, Table S5), indicating that G-type units were more prone to depolymerization and/or degradation during the pretreatment [50]. In addition to the degradation of the G-type units, the diminution of the G_2 and G_6 signals in $PEA_{0.4}$-L were also attributed to the condensation between the C-2 and C-6 of the G-type units and C-α in the S-type units [8,51].

The ^{31}P NMR results showed that the contents of hydroxyl groups in the separated lignin samples were clearly affected by the hydrogen bond acidity of PEA-DESs (Fig. S3). Based on the results in Fig. 6a, the contents of aliphatic and *p*-hydroxyphenyl hydroxyl groups generally decreased as the α values increased, while the C_5-substituted and guaiacyl OH contents decreased. The aliphatic and *p*-hydroxyphenyl OH contents were mildly decreased and converted to benzylic carbocation intermediates under the protective effect of the stabilizing reagent (EG). Therefore, aliphatic OH groups remained the primary OH groups in all lignin samples, which accounted for 61.9–80.9 % of the total OH groups. These results indicated that the lignin fractions were predominantly alkoxylated and less condensed, which was consistent with the HSQC NMR results. Besides, the content of total phenolic OH groups increased from 1.02 to 1.72mmol g^{-1} with the increase of α values from 1.17 to 1.75 (Table S6), which mainly originated from the increase in C_5-substituted OH content. In contrast, the content of G-type OH increased only slightly from 0.36 to 0.55mmol g^{-1}, indicating the formation of G-type lignin fragments containing *para*-phenolic OH after part of the cleaved β-aryl ether linkage preferentially participated in the lignin condensation [7]. Furthermore, the contents of aliphatic OH (Aliphatic OH content $= -4.35\alpha^2 + 10.09\alpha$-$1.55$, R^2: 0.99) and phenolic OH (Phenolic OH content $= 2.66\alpha^2$-$6.58\alpha + 5.09$, R^2: 0.98) showed strong quadratic dependencies with the α values of PEA-DESs (Fig. 6b), which is important for future tailoring of the reactivity of extracted lignin. The molecular weight distribution of fractionated lignin by different DESs is exhibited in Fig. 6c. The M_w decreased with the increasing α values and showed strong quadratic dependencies (Fig. 6d), which was different from the previously reported results for the M_w changes for γ-valerolactone-isolated lignin [43]. These results implied that the PEA-DES extracted lignin mainly underwent depolymerization rather than condensation during the lignin fractionation. Besides, the gradually decreased PDI values indicated that the condensation between the lignin fragments was inhibited by the incorporation of EG to form a more uniform lignin [52]. From these comprehensive results, it was clearly concluded that the structural properties of the extracted lignin fractions could be easily adjusted by tailoring the PEA-DES systems.

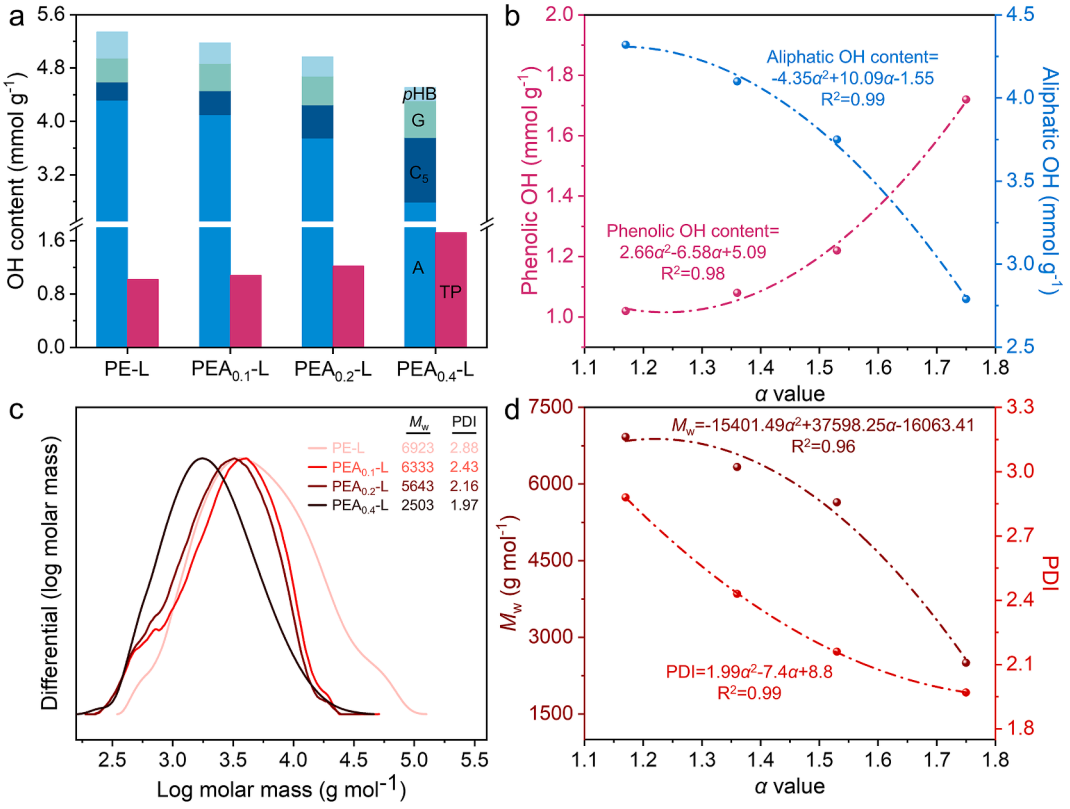


Fig. 6. Hydroxyl group contents (a) and the molecular weight distributions (c) of lignin fractionated by different PEA-DESs, and correlation between the α value of PEA-DESs and the content of phenolic and aliphatic OH (b), and M_w and PDI (d).

3.5. Fractionation and structural evolution mechanism of lignin by PEA-DESs pretreatment

The mechanism for fractionation and structural evolution of lignin during the PEA-DESs pretreatment is described in Fig. 7. It was well known that lignocellulose is is a remarkably complex assembly of cellulose, hemicellulose, and lignin [53]. During the pretreatment of lignocellulose with PEA-DESs, the added $AlCl_3$ enhanced the hydrogen-bond acidity and assisted the release of pyridine cations. The high hydrogen-bond acidity of PEA-DESs efficiently cleaved the benzyl ester and benzyl ether bonds in LCCs as well as the β-aryl ether bonds in lignin, leading to significant removal of hemicellulose and lignin [27,35]. Furthermore, pyridine cations with uneven electron density distribution have a strong electron-withdrawing nitrogen, which can strengthen the hydrogen bonding with aromatics and phenolics, affording a higher affinity to lignin and thus facilitating dissociation [24,25]. Meanwhile, the α-OH in the β-aryl ether linkages of lignin were protonated by H^+ and then dehydrated to form active benzyl carbocation, which were stabilized by EG to primarily form α-alkoxylated β-O-4 substructures and to secondarily generate a condensed structure with G-type aromatic units. It is worth noting that lignin condensation generally occurs on the active benzyl carbocation from S-type lignin. Additionally, weak depolymerization occurred during the efficient dissociation of lignin by PEA-DESs, allowing the generation of fractionated lignin with natural macromolecular structure.

3.6. Valorization of PEA DESs-extracted lignin

The relatively high content of β-aryl ether in the PEA-DESs extracted lignin makes it an attractive raw material for producing aromatic chemicals *via* depolymerization. However, most of the existing depolymerization methods convert protolignin, *i.e.,* lignin with α-OH groups in the β-O-4 linkages [12]. Lignin polyol, a typical protolignin derivative with α-alkoxylated β-O-4 structures, might have an impact on their special application. Usually, functionalization-defunctionalization is essential for functional material applications of lignin [54,55]. To make this protolignin derivative compatible with the developed valorization methods, we proposed taking advantage of the reversibility of the α-etherification to convert lignin polyol into native-like lignin. The de-etherification of $PEA_{0.2}$-L extracted from the poplar, pine, and bagasse was demonstrated by the HSQC NMR analysis (Fig. 8a-c). After de-etherification, the typical cross-peak of native β-O-4 linkage was significantly increased in the $PEA_{0.2}$-L from bagasse, and even reappeared in $PEA_{0.2}$-L extracted from poplar and pine. Specifically, the β-O-4 linkages in the $PEA_{0.2}$-L extracted from poplar, pine and bagasse increased from 0, 0, 5.5/100 Ar to 21.5, 25.7, 35.7/100 Ar, respectively. In all cases, substantially reduced peaks for the α-alkoxylated β-O-4 signals (A'_{α}, A'_{EG}) were observed, indicating that the lignin polyol derivative had been successfully transformed to protolignin. However, the total β-O-4 content in all the obtained lignin samples decreased to varied degrees after the de-etherification mediated by acid.

Hydrogenolysis has been considered as an important approach to convert lignin into aromatic compounds [6,56]. All the extracted $PEA_{0.2}$-L (from different lignocellulose) and the de-etherified $PEA_{0.2}$-L (De-$PEA_{0.2}$-L) samples exhibited well- retained β-O-4 structures, which was beneficial for achieving a high yield of aromatic compounds. The yields of various lignin-derived monomers obtained by hydrogenolysis are shown in Fig. 8d and Table S7. As reported, monomers **1**, **2**, **3**, and **4** were generally generated through the cleavage of β-O-4 linkages in lignin [57]. Because the contents of total β-O-4 linkages in CEL, $PEA_{0.2}$-L, and De-$PEA_{0.2}$-L from poplar and pine decreased, the yield of the targeted monomers decreased gradually, suggesting that the β-O-4 linkage content was the crucial factor affecting the yield of monophenols. In particular, the monomer yield from $PEA_{0.2}$-L-poplar with a β-O-4 linkage of 42.1/100 Ar was 27.5 %, which is roughly 7 times higher than that reported for other acidic DESs-extracted lignin [21]. Moreover, it was observed that the yields of monophenols from various $PEA_{0.2}$-L samples were only slightly lower than those from the corresponding CEL samples (Fig. 8d) because of the lower contents of β-O-4 linkages in $PEA_{0.2}$-L than that in CEL. Nevertheless, the product distribution was relatively uniform in both CEL and $PEA_{0.2}$-L, suggesting that PEA-DESs extracted lignin with well-preserved β-aryl ether bonds is a very promising alternative to protolignin. The bagasse lignin was composed of *p*-coumaric acid and ferulic acid pendant units, which were linked through both ester and ether linkages. These hydroxycinnamic acids underwent esterification with methanol followed by their subsequent hydrogenation to produce monomers **5** and **6** [58]. After PEA-DESs extraction and subsequent de-etherification, the hydroxycinnamic acid content was reduced, thus resulting in a significant decrease in the yields of **M5** and **M6**. Overall, the lignin extracted by PEA-DESs

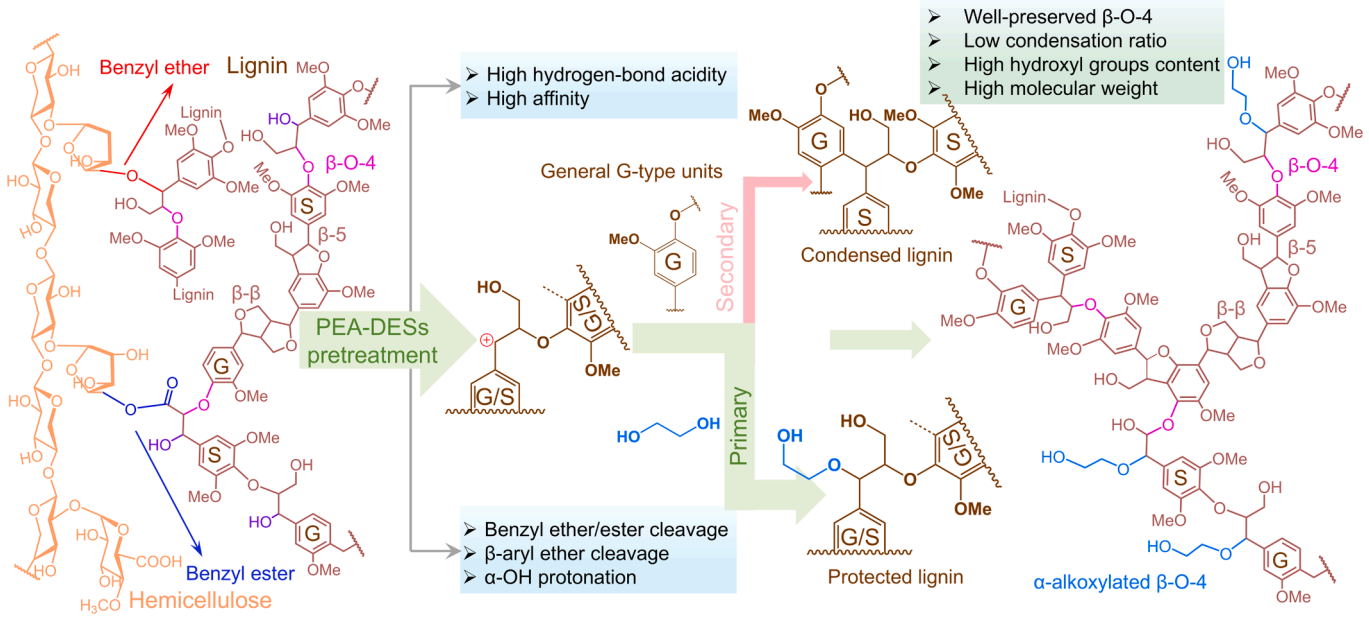


Fig. 7. Proposed processes of chemical structure evolution of native lignin during the PEA-DESs pretreatment.

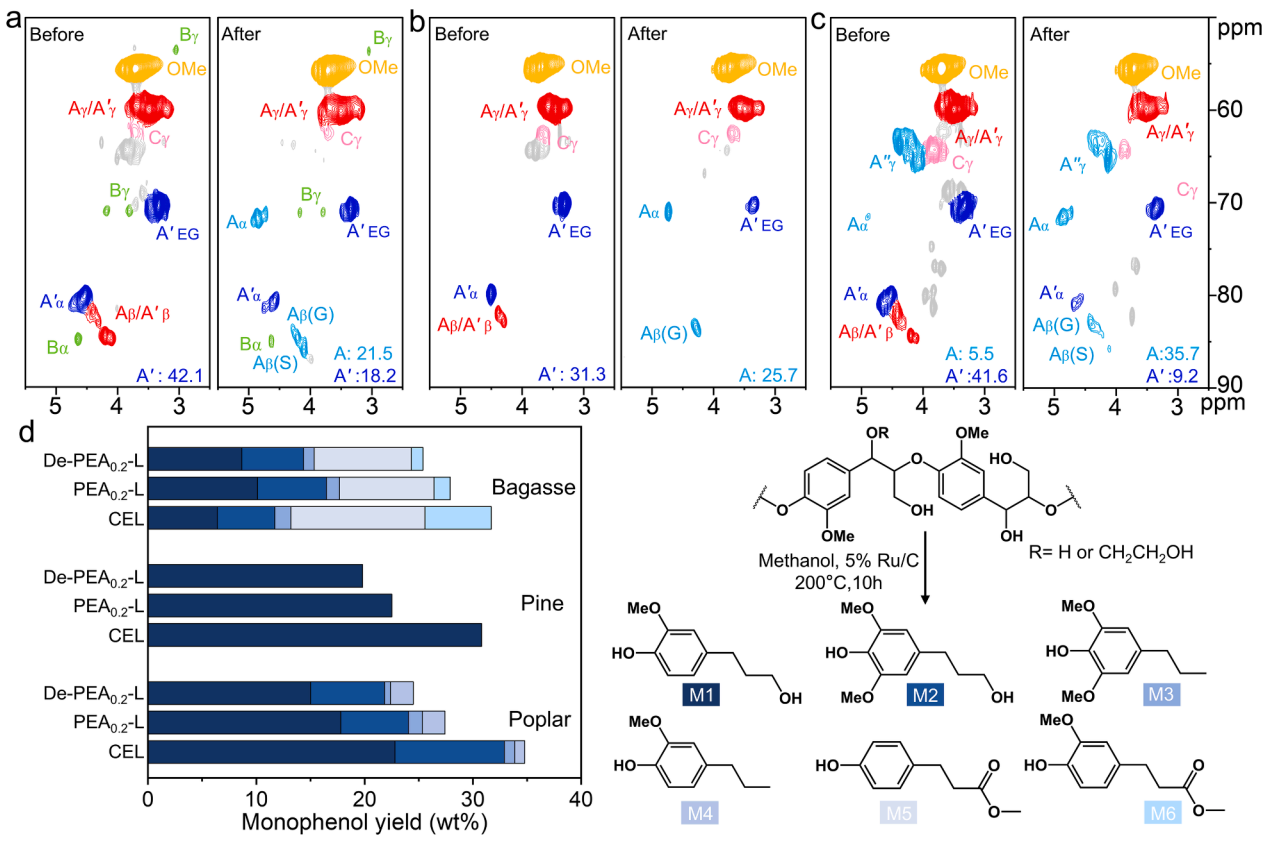


Fig. 8. 2D HSQC NMR analysis of $PEA_{0.2}$-L fractionated from different lignocellulose before and after de-etherification (a: poplar, b: pine, c: bagasse), and the monophenol yields after catalytic hydrolysis of CEL, $PEA_{0.2}$-L, and de-etherized $PEA_{0.2}$-L from different lignocelluloses (d).

fractionation well preserved β-O-4 linkages, thereby resulting in high yields of the aromatic monomers. More importantly, this developed PEA-DES-pretreatment approach provided an effective stabilization method to achieve reactive lignin samples for the production of valuable lignin-based aromatic monomers.

3.7. *Enzymolysis of cellulose-rich residues*

One of the main purposes of lignocellulose pretreatment was to generate cellulose-rich residues, which could be used as the raw material to produce sugars *via* saccharification. In this study, the developed PEA-DESs demonstrated good ability for lignocellulose deconstruction and delignification under mild conditions, and the cellulose retention in the residues exceeded 90 %, which facilitated a high sugar yield by enzymatic saccharification. Among the cellulose-rich residues obtained from the poplar pretreatment, the residue from the PEA-DESs(1:4:0.2) pretreatment system at 70 °C provided the highest fermentable sugar yield (88.2 % glucose) after the enzymolysis of the cellulose-rich residue for 72 h. This yield was 4.6 times higher than that released from the raw poplar (Fig. S4a), which was mainly attributed to efficient removal of lignin and hemicellulose by the developed PEA-DESs, thus being beneficial for the interaction between cellulose and enzymes. However, further removal of lignin and hemicellulose reduced the glucose yield. The residue from the PEA-DESs-(1:4:0.4) pretreatment system exhibited lower enzymatic hydrolysis activity than that from the PEA-DESs-(1:4:0.2) pretreatment system, but a high glucose yield (81.5 %) was still achieved. The lower glucose yield from the PEA-DESs-(1:4:0.4)-generated residue was mainly caused by the lignin structure in the generated cellulose-rich residues. It was demonstrated that the hydrophobic interaction between the lignin in cellulose-rich residue and cellulase showed a long-range attraction (the main driver for the nonproductive adsorption behavior of cellulase on lignin), especially for the condensed lignin with stronger hydrophobicity [59]. As discussed above, more acidic PEA-DESs-(1:4:0.4) caused greater lignin condensation in the cellulose-rich residue, enhancing the lignin-cellulase hydrophobic interactions, which resulted in a lower enzymatic activity than that of the PEA-DESs-(1:4:0.2)-generated residue. In addition, the bagasse residue obtained from the PEA-DESs-(1:4:0.2) system gave an extremely high glucose yield of 95.6 % after the same enzymatic hydrolysis (Fig. S4b). In conclusion, the EG was grafted onto the α-position of the lignin side chain to form α-etherified lignin polyol, which effectively constrained the lignin condensation and preserved hydrophilicity of the lignin side chain region, thus limiting nonproductive lignin-enzyme interactions. This was the primary reason for the high sugar yield from the cellulose-rich residues obtained in the proposed PEA-DESs pretreatment system, this holds promise for the development of economically viable and cost-effective enzyme-mediated biorefinery processes.

4. Conclusions

In summary, reactive PEA-DESs were developed to fractionate lignocellulose, eventually resulting in efficient co-valorization of lignin and polysaccharide fractions. PEA-DESs were able to mildly and efficiently fractionate lignin with natural structures (well-preserved β-O-4 linkage, abundant hydroxyl groups, and high molecular weight) from lignocelluloses (poplar, pine, and bagasse) by the synergistic effect of high hydrogen-bond acidity and high lignin affinity. During the PEA-DESs pretreatment, lignin was efficiently dissociated and modified by ethylene glycol through α-alkoxylation that facilitated high retention of β-O-4 linkages and markedly blocked lignin recondensation, contributing to the high yield of aromatic monomers. In addition, hydroxyl-rich lignin decreased the nonproductive adsorption of cellulase, thus promoting the enzymatic hydrolysis of cellulose-rich residues to generate

high yields of sugar. Notably, the evolution of lignin structure under PEA-DESs pretreatment was comprehensively investigated, which provided a basis for understanding lignin dissociation and obtaining lignin with a controlled structure. We believe that this fractionation system offers great promise for the efficient extraction of lignin with controllable structure and high-value utilization of lignocellulose components.

CRediT authorship contribution statement

Yang Wang: Writing – original draft, Methodology, Investigation. **Qiaoling Liu:** Methodology, Investigation. **Chuanyu Yan:** Methodology, Formal analysis. **Guoyong Song:** Methodology, Funding acquisition, Formal analysis. **William S. Price:** Writing – review & editing, Methodology, Formal analysis. **Gang Zheng:** Data curation. **Allan M. Torres:** Methodology, Data curation. **Zhimin Xue:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2024.158825.

Data availability

Data will be made available on request.

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